

Linear Algebra

- Midterm: 35
End term: 45
Quiz: 10 + 10.

Linear functional

$$f: V \rightarrow \mathbb{F} \quad \text{L.T.}$$

$V' = \mathcal{L}(V, \mathbb{F})$ set of all ^{linear} functionals on V

↑ dual space of V (dual of V)

If $V \leftarrow f.d.$ & $n = \dim(V)$ then

$$\dim(V) = n = \dim(V').$$

If $f_1 = f_2$ then $f_1(x) = f_2(x) \iff \forall x \in V$

$$f_1(x_1) = f_1(x_2) \iff x_1 = x_2 \iff \forall f \in V' \iff x_1 = x_2$$

→ $(\Rightarrow)$ Suppose $x_1 \neq x_2 \Rightarrow x_1 - x_2 \neq 0$.

$$\exists f \in V' \text{ s.t. } f(x_1 - x_2) \neq 0. \quad (\text{can do this but why?})$$

$$\Rightarrow f(x_1) \neq f(x_2)$$

extend basis and generic def. logic

Let $\dim(V) = n$.

Basis $\{v_1, \dots, v_n\}$.

→ for each $i = 1, \dots, n \exists! f_i \in V'$ s.t. $f_i(v_j) = \delta_{ij}$.

$$v \in V \Rightarrow v = \sum_{i=1}^n \alpha_i v_i \Rightarrow f_i(v) = \alpha_i.$$

$\{f_i\}_{i=1}^n$ are coordinate fⁿ for basis $\{v_1, \dots, v_n\}$

V	W
$v_1, \dots, v_n$ basis	$w_1, \dots, w_n$ any n vector
$\exists! T: V \rightarrow W \in \mathcal{L}(V, W)$ s.t. $T(v_i) = w_i$	

claim: $\{f_1, \dots, f_n\}$ is L.T.

$$\alpha_1 f_1 + \dots + \alpha_n f_n = 0.$$

$$\Rightarrow (\alpha_1 f_1 + \dots + \alpha_n f_n)(v_i) = 0 \quad (v_i) = 0 \Rightarrow \alpha_i = 0 \quad \forall i \quad \square$$

So $\{f_i\}_{i=1}^n$ form basis of V' .

Dual Basis: Let V be a f.d.v.s. & $v_1, \dots, v_n$ be basis of V . then basis $\{f_1, \dots, f_n\}$ is called dual basis of V' if $f_i(v_j) = \delta_{ij}$.

Obs. V be v.s. $\{v_1, \dots, v_n\} \in \text{BASIS}$.
 $\{f_1, \dots, f_n\} \rightarrow \text{Dual Basis}$.

$$f \in V' \Rightarrow f = \sum_{i=1}^n f(v_i) f_i$$

$$v \in V \Rightarrow v = \sum_{i=1}^n f_i(v) v_i$$

$$f = \sum_{i=1}^n \alpha_i f_i \Rightarrow f(v) = \sum_{i=1}^n \alpha_i f_i(v)$$

$$f(v_j) = \sum_{i=1}^n \alpha_i f_i(v_j) = \alpha_j \quad \forall j$$

$$\Rightarrow f = \sum_{i=1}^n f(v_i) f_i$$

Obs. $V \mapsto$ f.d.v.s.

$\{v_1, \dots, v_n\}$ basis.

Def $(a_1, \dots, a_n) \in \mathbb{F}^n$

Then $g \in V'$.

$$g: V \rightarrow \mathbb{F}$$

$$g(v) = g(\sum_i \alpha_i v_i) = \sum_i \alpha_i g(v_i) = \sum_i \alpha_i a_i \leftarrow \text{define.}$$

Conversely $g \in V'$. $g(\sum_i \alpha_i v_i) = \sum_i \alpha_i \underbrace{g(v_i)}_{a_i}$.

Example: Let x_1, x_2, x_3 distinct real #s.

$$V = \mathcal{P}_2(\mathbb{R}).$$

for $i=1, 2, 3$, then $L_i: \mathcal{P}_2(\mathbb{R}) \rightarrow \mathbb{R}$

$$L_i(p) = p(x_i).$$

then $L_i \in V'$ for $i=1, 2, 3$.

$$\alpha_1 L_1 + \alpha_2 L_2 + \alpha_3 L_3 = 0.$$

$$\Rightarrow \alpha_1 p(t_1) + \alpha_2 p(t_2) + \alpha_3 p(t_3) = 0 \quad \forall p \in \mathcal{P}_2(\mathbb{R}).$$

$$\Rightarrow \begin{bmatrix} 1 & 1 & 1 \\ t_1 & t_2 & t_3 \\ t_1^2 & t_2^2 & t_3^2 \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

$$\det = t_2 t_3^2 - t_2^2 t_3 - t_1 t_3^2 + t_1^2 t_3 + t_1 t_2^2 - t_1^2 t_2$$

$\neq 0$ (How? use t_1, t_2, t_3 distinct)
 L_1, L_2, L_3 is a basis of V' .

Can we find a basis B of V ^{in above example.} s.t. B' is the dual of B .
 i.e., find q_1, q_2, q_3 s.t. $L_i(q_j) = \delta_{ij}$

$$L_i(q_j) = q_j(t_i) = \delta_{ij}$$

$$q_1 = (x - t_2)(x - t_3) / (t_1 - t_2)(t_1 - t_3)$$

$$q_2 = (x - t_1)(x - t_3) / (t_2 - t_1)(t_2 - t_3)$$

$$q_3 = (x - t_1)(x - t_2) / (t_3 - t_1)(t_3 - t_2)$$

Q: For a given basis B' of V' can we find a B of V s.t. B' is dual basis of B ?

Obs. $v \in V$, $L_v : V' \rightarrow \mathbb{F}$.

$$L_v(f) = f(v)$$

check $L_v \in V''$.

claim: $T: V \rightarrow V''$, $T(v) = L_v$ is an isomorphism.

well defined.

$$u = v$$

$$f(u) = f(v) \quad \forall f \in V'$$

$$\Rightarrow L_u(f) = L_v(f) \quad \forall f \in V'$$

$$\Rightarrow L_u = L_v$$

$$\Rightarrow T(u) = T(v).$$

check that T is an isomorphism (linear, one-one, onto).

obs. if $L \in V'' \Rightarrow \exists v \in V$ s.t. $T(v) = L \Rightarrow L_v = L$.

Recall: Let V be f.d.v.s. Each Basis of V' is the dual basis of some basis for V .

Let $\{f_1, \dots, f_n\}$ be basis of V' . $\exists \{L_1, \dots, L_n\}$ dual basis of V' . $L_i(f_j) = \delta_{ij}$.

$\Rightarrow \exists \{v_1, \dots, v_n\} \in V$ s.t. $L_i = L_{v_i}$

we need to show $\{v_1, \dots, v_n\}$ is a basis and $f_i(v_j) = \delta_{ij}$

$$\begin{cases} f_i(v_j) = L_{v_j}(f_i) = L_j(f_i) = \delta_{ji} = \delta_{ij} \\ \alpha_1 v_1 + \dots + \alpha_n v_n = 0 \\ f_i(\alpha_1 v_1 + \dots + \alpha_n v_n) = f_i(0) = 0 \Rightarrow \alpha_i = 0 \quad \forall i \end{cases}$$

Hyperspace: A maximal proper subspace of V is called hyperspace.

W be maximal proper subspace of $V \Rightarrow \forall U \subseteq V; W \subseteq U$
then $W = U$ or $U = V$.

For f.d. case. $\dim(V) = n$, any subspace of $\dim = n-1$ is a hyperspace.

obs: Let V be f.d.v.s. Each hyperspace is a nullspace of some linear functional on V .

Proof: Let $f \in V'$ by rank nullity thm.
then $\dim(\text{null}(f)) = n-1 \Rightarrow \text{null}(f)$ is a hyperplane

Let W be a hyperspace. We want $f \in V'$ s.t. $\text{null}(f) = W$

Since W is hyperspace $\Rightarrow \dim(W) = n-1$

Let $\{w_1, \dots, w_{n-1}\}$ basis of W .

Extend them $\{w_1, \dots, w_{n-1}, 0\}$ basis of V .

dual basis $\{f_1, \dots, f_n\}$ then $\text{null}(f_n) = \text{span}\{w_1, \dots, w_{n-1}\} = W$.

(ABOVE RESULTS ALSO TRUE FOR INFINITE DIM CASE)

Let $W \subseteq V$ and $\dim(W) = n-2$.

then $\exists f_1, f_2 \in V'$ s.t. $W = \text{null}(f_1) \cap \text{null}(f_2)$.

Proof: $\{w_1, \dots, w_{n-2}\}$ basis of W .

$\{w_1, \dots, w_{n-2}, v_1, v_2\}$ basis of V .

$$f_1: V \rightarrow \mathbb{F}, \quad f_1(w_1) = f_1(w_2) = \dots = f_1(v_2) = 0, f_1(v_1) = 1$$

$$f_2: V \rightarrow \mathbb{F}, \quad f_2(w_1) = \dots = f_2(v_1) = 0, f_2(v_2) = 1$$

$$\begin{aligned} \text{null}(f_1) &= \langle \{w_1, \dots, w_{n-2}, v_2\} \rangle \\ \text{null}(f_2) &= \langle \{w_1, \dots, w_{n-2}, v_1\} \rangle \end{aligned} \Rightarrow \text{null}(f_1) \cap \text{null}(f_2) = W,$$

obs.: Each d -dimensional subspace of V is the intersection of null space of $(n-d)$ linear functional of V .

Annihilator of Set Let V be V.S., $S \subseteq V$ subset.

$$S^\circ = \{f \in V' \mid f(v) = 0 \forall v \in S\}$$

S° is a subspace of V' . (Exercise)

$$\{0\}^\circ = V', \quad \{V\}^\circ = \{0\}.$$

Result: $V \leftarrow$ s.d.v.s.

$W \subseteq V$ subspace

$$\dim(V) = \dim(W) + \dim(W^\circ).$$

Proof: $\{w_1, \dots, w_d\}$ basis of W .

$\{w_1, \dots, w_d, v_1, \dots, v_{n-d}\}$ basis of V .

$\Downarrow$ dual

$\{f_1, \dots, f_d, f_{d+1}, \dots, f_n\}$ dual basis of V .

$$\text{Claim: } W^\circ = \langle \{f_{d+1}, \dots, f_n\} \rangle. \quad (\text{Ex.})$$

Exercise ① $f_1(x_1, x_2, x_3, x_4) = x_1 + 2x_2 + 2x_3 + x_4$
 $f_2 = 2x_2 + x_4$
 $f_3 = -2x_1 - 4x_3 + 3x_4$.

find subspace of $\mathbb{R}^4$ which is annihilated by f_1, f_2, f_3 .

Exercise ② W be subspace spanned by $w_1 = (2, -2, 3, 4, -1)$
 $w_2 = (-1, 1, 2, 5, 2)$
 $w_3 = (0, 0, -1, -2, 3)$
 $w_4 = (1, -1, 2, 3, 0)$

find W° and a basis for W°

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Recall: $v \in V$

$$L_v : V' \rightarrow \mathbb{F}, \quad L_v(f) = f(v).$$

$$T: V \rightarrow V''; \quad T(v) = L_v \quad \text{isomorphism.}$$

$$V \longrightarrow V' \longrightarrow V'', \quad V \cong V''$$

So we may say V' and V dual of each other.

$$S \subseteq V, \quad S^\circ = \{ f \in V' \mid f(v) = 0 \quad \forall v \in S \}.$$

$$S \subseteq V', \quad S^\circ = \{ L \in V'' \mid L(f) = 0 \quad \forall f \in S \}.$$

$$= \{ L_v \in V'' \mid \exists v \in V, L_v(f) = 0 \quad \forall f \in S \}$$

$$= \{ v \in V \mid f(v) = 0 \quad \forall f \in S \}.$$

$\leadsto$ as $V'' \cong V$, so subset of V

if $\uparrow$ just a set
 $W \subseteq V$

$$W^\circ = \{ f \in V' \mid f(v) = 0 \quad \forall v \in W \}.$$

subspaces $\nwarrow$ $W^{\circ\circ} = \{ v \in V \mid f(v) = 0 \quad \forall f \in W^\circ \}.$

$$\text{and } W \subseteq W^{\circ\circ}.$$

Result: V f.d.v.s., $W \subseteq V$ subset, $\text{span}(W) = W^{\circ\circ}$.

clearly $\text{span}(W) \subseteq W^{\circ\circ}$. $\longrightarrow$ (*)

$$\dim(V)^S = \dim(S) + \dim(S^\circ)$$

$$\dim(V') = \dim(S^\circ) + \dim(S^{\circ\circ})$$

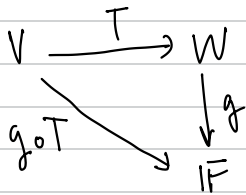
$$\dim(V) = \dim(V') \Rightarrow \dim(S) = \dim(S^{\circ\circ}) \longrightarrow (**)$$

by $\circledast$ & $\circledast\circledast$ result follows.

Transpose of a L.T.

$V, W \leftarrow \text{V.S.}$

$T: V \rightarrow W$ L.T.



$\exists$ a unique L.T. $T^t: W' \rightarrow V'$

$$T^t(g) = g \circ T.$$

Exercise: V, W be V.S. B_V, B_W basis resp.
 $T: V \rightarrow W$ be linear transformation.
 B'_V, B'_W dual basis of B_V & B_W .

$$[T^t]_{B'_W}^{B'_V} = \left([T]_{B_W}^{B_V} \right)^t \quad \text{matrix transpose.}$$

Result: Let V, W be v.s.

$T: V \rightarrow W$ be L.T.

$$\text{then } [\text{null}(T^t)] = [\text{range}(T)]^\circ$$

If $V, W = \text{F.d.v.s.}$

- ① $\text{rank}(T) = \text{rank}(T^t)$
- ② $\text{range}(T^t) = [\text{null}(T)]^\circ$.

$$\text{Let } g \in \text{null}(T^t) \Leftrightarrow T^t(g) = g \circ T = 0$$

$$\begin{aligned} \Leftrightarrow g \circ T(v) &= 0 \quad \forall v \in V \\ \Leftrightarrow g(T(v)) &= 0 \quad \forall v \in V \\ \Leftrightarrow g &\in [\text{Range}(T)]^\circ. \end{aligned}$$

$$\dim(\text{Range}(T)) + \dim([\text{Range}(T)]^\circ) = \dim(W)$$

$$\dim(\text{Range}(T)) = \dim(W) - \dim([\text{Range}(T)]^\circ)$$

$$\dim(\text{Range}(T)) = \dim(W) - \dim(\text{Null}(T^t))$$

$$\dim(\text{Range}(T)) = \dim(W') - \dim(\text{Null}(T^t))$$

$$\dim(\text{Range}(T)) = \dim(\text{Range}(T^t)).$$

$$\text{exercise: } \text{range}(T^t) = [\text{null}(T)]^\circ$$

Let $A_{m \times n}$ matrix over $\mathbb{F} = \mathbb{R} \text{ or } \mathbb{C}$.

$$\text{column rank}(A) = \text{Row rank}(A^t).$$

Proof: $T: \mathbb{F}^n \rightarrow \mathbb{F}^m; T(x) = Ax$

$$\begin{array}{c} \swarrow \quad \searrow \\ \text{basis } (e_1, \dots, e_n)_{B_1} \quad (e_1, \dots, e_m)_{B_2} \end{array}$$

$$[T]_{B_2}^{B_1} = A. \quad A = [a_1 \dots a_n], \quad x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$$

$$\text{rank}(T) = \text{column rank}(A)$$

$$B_1' = \{e_1, \dots, e_n\}, \quad B_2' = \{e_1, \dots, e_m\}.$$

$$[T^t]_{B_2'}^{B_1'} = A^t. \quad \text{rank}(T^t) = \text{column rank}(A^t) = \text{row rank}(A)$$

$$\text{rank}(T) = \text{rank}(T^t)$$

$$\Rightarrow \text{column rank}(A) = \text{row rank}(A^t).$$

Exercises:

$V \in \text{f.d.v.s.}$

W_1, W_2 subspaces of V .

$$(1) (W_1 \cap W_2)^\circ = W_1^\circ + W_2^\circ$$

$$(2) (W_1 + W_2)^\circ = W_1^\circ \cap W_2^\circ$$

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Let $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{F}$ be a function.

$\langle \cdot, \cdot \rangle$ is called an inner product on V if:

$$(1) \langle x, x \rangle \geq 0 \quad \forall x \in V$$

$$(2) \langle x, x \rangle = 0 \iff x = 0.$$

$$(3) \langle x, y \rangle = \overline{\langle y, x \rangle}$$

$$(4) \langle \alpha x + y, z \rangle = \alpha \langle x, z \rangle + \langle y, z \rangle. \quad \forall \alpha \in \mathbb{F}, x, y, z \in V.$$

e.g.: Suppose $\exists f: V \times V \rightarrow \mathbb{C}$

$$x \neq 0, x \in V, f(x, x) > 0$$

$$ix \neq 0, ix \in V, f(ix, ix) > 0.$$

$$f(ix, ix) = i^2 f(x, x) = -f(x, x) < 0$$

So (iii) not property needed

$$\text{eg. (1) } \mathbb{R}^n \quad \langle x, y \rangle = \sum_{i=1}^n x_i y_i$$

$$(2) \mathbb{C}^n \quad \langle x, y \rangle = \sum_{i=1}^n x_i \bar{y}_i$$

$$\begin{aligned} f(ix, ix) &= if(x, ix) \\ &= i(-i)f(x, x) \\ &= 1 \cdot f(x, x) > 0 \end{aligned}$$

$$\textcircled{3} \quad V = M_{n \times n}(\mathbb{F})$$

$$\langle A, B \rangle = \text{trace}(AB^*)$$

$$\textcircled{4} \quad V = C[0,1]$$

$$\langle f, g \rangle = \int_0^1 f(t) \overline{g(t)} dt.$$

Result: V and W be v.s. over $\mathbb{F}$.

$$\hookrightarrow \langle \cdot, \cdot \rangle_W$$

$$T: V \rightarrow W \quad \text{linear one-to-one}$$

$$\langle \cdot, \cdot \rangle: V \times V \rightarrow \mathbb{F}$$

$$\langle u, v \rangle = \langle T(u), T(v) \rangle_W \quad \text{is an i.p. on } V.$$

e.g. V is vector space over $\mathbb{R}$ and $\dim(V) = n$.

as $V \cong \mathbb{R}^n$ so $\exists \psi: V \rightarrow \mathbb{R}^n$ isomorphism

then $\langle u, v \rangle_V := \langle \psi(u), \psi(v) \rangle_{\mathbb{R}^n}$ is i.p. on V .

e.g. $V = C[0,1], W = V$

$$T: V \rightarrow W.$$

$$T(f(t)) = t \cdot f(t). \quad \leadsto \text{linear transform one-to-one (bijective)}.$$

$$\langle f, g \rangle = \langle tF(t), tG(t) \rangle = \int_0^1 t^2 f(t) \overline{g(t)} dt.$$

V i.p.s. $S \subseteq V$ is orthogonal if

$$\langle x, y \rangle = 0 \quad \forall x \neq y, x, y \in S.$$

$$\text{length}(x) = \sqrt{\langle x, x \rangle} = \|x\|.$$

$$\text{angle b/w } x \text{ \& } y = \frac{\langle x, y \rangle}{\|x\| \cdot \|y\|}$$

orthonormal: $S \subseteq V, \langle x, y \rangle = 0 \quad \forall x \neq y, x, y \in S$
 $\|x\| = 1 \quad \forall x \in S.$

u, v be 2 vectors in i.p.s. V .

can we write $u = \alpha v + w$ s.t. $w \perp v$
 $w = u - \alpha v$

$$0 = \langle w, v \rangle = \langle u - \alpha v, v \rangle = \langle u, v \rangle - \alpha \langle v, v \rangle$$

$$\Rightarrow \alpha = \frac{\langle u, v \rangle}{\langle v, v \rangle}.$$

$$u = \underbrace{\frac{\langle u, v \rangle}{\langle v, v \rangle} v}_{\alpha v} + \underbrace{u - \frac{\langle u, v \rangle}{\langle v, v \rangle} v}_w, \quad v \neq 0.$$

Cauchy-Schwarz Ineq.: $u, v \in V \leftarrow$ i.p.s.

$$|\langle u, v \rangle| \leq \|u\| \cdot \|v\| \quad \text{--- } (*)$$

equality holds in $(*)$ iff $\{u, v\}$ is linearly dependent.

11-08-2026

Gram-Schmidt orthogonalisation.

$V \leftarrow$ i.p.s
 $\{v_1, \dots, v_n\}$ L.I.

then $\exists \{q_1, \dots, q_n\}$ orthonormal s.t.

$$\text{Span} \{v_1, \dots, v_k\} = \text{Span} \{q_1, \dots, q_k\} \quad \forall k=1, \dots, n$$

Idea: $\tilde{q}_1 = v_1, \quad q_1 = \tilde{q}_1 / \|\tilde{q}_1\|.$

$$\tilde{q}_2 = v_2 - \langle v_2, q_1 \rangle q_1, \quad q_2 = \tilde{q}_2 / \|\tilde{q}_2\|.$$

$$\tilde{q}_3 = v_3 - \langle v_3, q_1 \rangle q_1 - \langle v_3, q_2 \rangle q_2, \quad q_3 = \tilde{q}_3 / \|\tilde{q}_3\|.$$

$$\vdots$$
$$\tilde{q}_k = v_k - \sum_{i=1}^{k-1} \langle v_k, q_i \rangle q_i, \quad q_k = \tilde{q}_k / \|\tilde{q}_k\|.$$

$\forall$ orthonormal basis, $v = \sum_{i=1}^n \langle v, q_i \rangle q_i.$

QR factorization (Teaser): $A_{n \times n}$ non singular.

$$A = \underbrace{Q}_{\substack{\text{unitary} \\ n \times n}} \underbrace{R}_{\substack{\text{upper } \Delta \\ n \times n}}$$

$$A_{n \times m} \quad \text{rank}(A) = r$$

$$A = \underbrace{Q}_{\substack{n \times r \\ \text{orthonormal} \\ \text{columns}}} \underbrace{R}_{\substack{r \times m \\ \text{upper } \Delta}}$$

$$B = [b_1 | b_2 | \dots | b_n] \quad b_i's \text{ L.I as non singular.}$$

↓ G.S.

$$Q = [q_1 | q_2 | \dots | q_n]$$

$$r_{ij} = \begin{cases} \frac{\langle b_j, q_i \rangle}{\langle q_i, q_i \rangle} & i < j \\ \|q_i\| & i = j \end{cases}$$

$$R = (r_{ij})$$

$$A = QR.$$

Orthogonal complement: $V \leftarrow \text{i.p.s.}$, $U \subseteq V$ subspace

$$U^\perp = \{ v \in V \mid \langle v, u \rangle = 0, u \in U \}$$

$$\left. \begin{array}{l} V \leftarrow \text{i.p.s.} \\ U \leftarrow \text{f.d. subspace} \end{array} \right\} V = U \oplus U^\perp.$$

↳ subspaces too

Proof: $k = \dim(U)$, $\{q_1, \dots, q_k\}$ basis of U .

$$\text{claim: } V = U + U^\perp, \quad U \cap U^\perp = \{0\}.$$

$V \supseteq U + U^\perp$ is trivial.

$$v \in V \Rightarrow u = \sum_{i=1}^k \langle v, q_i \rangle q_i \in U$$

$$\Rightarrow v - u \in U^\perp \Rightarrow v = \underbrace{u}_U + \underbrace{v-u}_{\in U^\perp}$$

Exercise

$V \leftarrow$ f.d. i.p.s., $U \subseteq V$ subspace

$$U^{\perp\perp} = U.$$

Orthogonal Projection $V \leftarrow$ V.S.

$T: V \rightarrow V$ is projection if $T^2 = T$.

T is an orthogonal projection if:

① T is a projection: $T^2 = T$.

② $[\text{Range}(T)]^{\perp} = \text{Null}(T)$.

E.g.: $V \leftarrow$ i.p.s.
 $U \subseteq V$ subspace f.d.

Let $\{z_1, \dots, z_n\}$ orthonormal basis of U .

$$P_U: V \rightarrow V$$

$$P_U(v) = \sum_{i=1}^n \langle v, z_i \rangle z_i.$$

Exercise: check that: P_U is a projection $P_U^2 = P_U$.
 $[\text{Range } P_U]^{\perp} = \text{null}(P_U)$

Result: $V \leftarrow$ i.p.s.
 $U \leftarrow$ f.d. subspace.

Let $v \in V$

$$\|v - u^*\| = \min_{u \in U} \|v - u\|.$$

$\downarrow$
 $P_U(v)$

Su M Tu W Th F Sa
X L U U

$$\begin{aligned} \text{Proof: } \|v - P_U(v)\|^2 &\leq \underbrace{\|v - P_U(v)\|^2}_{\in U^{\perp}} + \underbrace{\| -w + P_U(v) \|^2}_{\in U} \quad \forall w \in U \\ &= \|v - w\|^2 \end{aligned}$$

$$\Rightarrow \|v - P_U(v)\| \leq \|v - w\| \quad \forall w \in U.$$

e.g.: Find the best approximation of $\sin x$ by a polynomial of deg 5. in $[-\pi, \pi]$.

From calculus: $\sin x \approx x - \frac{x^3}{3!} + \frac{x^5}{5!}$

from Lin Al: $V = C[-\pi, \pi]$, $\langle f, g \rangle = \int_{-\pi}^{\pi} f(x)g(x) dx$.
over $\mathbb{R}$

$$v = \sin x$$

$V \leftarrow$ set of all poly of deg ≤ 5 .

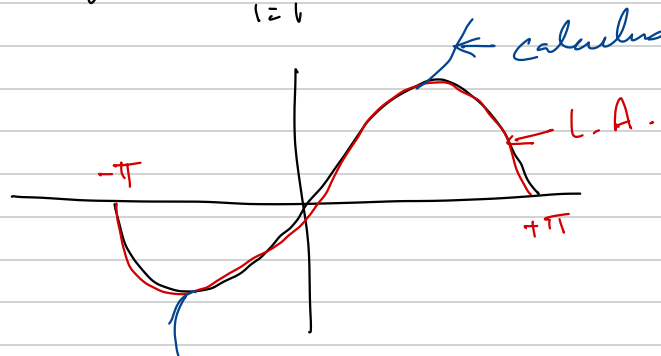
$$P_V(v) = ?$$

$$\text{basis} = \{1, x, x^2, x^3, x^4, x^5\}$$

$\downarrow$ G.S.

$$\text{orthonormal basis} = \{\sigma_1, \sigma_2, \sigma_3, \sigma_4, \sigma_5\}$$

$$P_V(v) = \sum_{i=1}^5 \langle \sin x, \sigma_i \rangle \sigma_i.$$



12-08-2026

Adjoint of an operator

let $V \leftarrow$ i.p.s

$$u \in V$$

$$f_u: V \rightarrow \mathbb{F}$$

$$f_u(v) = \langle v, u \rangle \quad \forall v \in V$$

$$\Rightarrow f_u \in V'$$

Converse: for a given $f \in V'$ does $\exists$ a $u \in V$ s.t. $f(v) = \langle v, u \rangle \quad \forall v \in V$?

Answer: No, if V is i.d.i.p.s.
Yes, if V is f.d.i.p.s.

Example: $V = \mathcal{P}(\mathbb{C})$ all poly. over $\mathbb{C}$.

$$L: V \rightarrow \mathbb{F}, \quad z \in \mathbb{C} \setminus \{0\}.$$

$$L(p(t)) = p(z). \quad \text{then } L \in V'.$$

$$\nexists q(t) \in V \text{ s.t. } L(p(t)) = \langle q(t), p(t) \rangle = \int_0^1 p(t) \overline{q(t)} dt$$

$$\text{if suppose } \exists q(t) \in V \Rightarrow L(p(t)) = \int_0^1 p(t) \overline{q(t)} dt = p(z) \quad \forall p \in V.$$

$$\text{take } r(t) = t - z.$$

observe that

$$\text{for any } h(t) \in V, \quad h_r(z) = h(z) - z h'(z) = 0$$

$$L(p r(t)) = \int_0^1 p(t) r(t) \overline{q(t)} dt = p r(z) = p(z) r(z) = 0.$$

$$\int_0^1 p(t) r(t) \overline{q(t)} dt = 0 \quad \forall p \in V$$
$$p(t) \overline{q(t)} r(t)$$

$$\Rightarrow \int_0^1 |q(t)|^2 |r(t)|^2 dt = 0$$

$$\Rightarrow |q(t)| |r(t)| = 0 \Rightarrow q(t) r(t) = 0 \quad \forall t$$

Now suppose V is f.d.v.s.

$\{v_1, \dots, v_n\} \leftarrow$ orthonormal basis

$$\text{Let } f \in V', \quad v \in V \Rightarrow v = \sum_{i=1}^n \langle v, v_i \rangle v_i$$

$$f(v) = \sum_{i=1}^n \langle v, v_i \rangle f(v_i)$$

$$f(v) = \sum_{i=1}^n \langle v, \overline{f(v_i)} v_i \rangle$$

$$f(v) = \left\langle v, \underbrace{\sum_{i=1}^n \overline{f(v_i)} v_i}_{u_f} \right\rangle.$$

$$\exists u_f \in V \text{ s.t. } f(v) = \langle v, u_f \rangle \quad \forall v \in V.$$

uniqueness: Let $\langle v, u_f \rangle = \langle v, w_f \rangle \quad \forall v \in V.$

$$\Rightarrow \langle v, u_f - w_f \rangle = 0 \quad \forall v \in V$$

$$\Rightarrow \langle u_f - w_f, u_f - w_f \rangle = 0$$

$$\Rightarrow u_f = w_f.$$

Conclusion: if V is f.d.i.p.s then $f \in V' \Leftrightarrow \exists$ unique $u_f \in V$
s.t. $f(v) = \langle v, u_f \rangle \forall v \in V$.

Adjoint of an operator

Let $V \leftarrow$ i.p.s.

$$T: V \rightarrow V \quad \text{L.T.}$$

then T has an adjoint on V if

$$\exists S \in L(V) \quad \text{s.t.}$$

$$\langle T(u), v \rangle = \langle u, S(v) \rangle \quad \forall u, v \in V.$$

S is called adjoint of T .

if adjoint exists it is unique.

Let S_1 and S_2 be 2 op. s.t.

$$\langle T(u), v \rangle = \langle u, S_1(v) \rangle \quad \forall u, v$$

$$\langle T(u), v \rangle = \langle u, S_2(v) \rangle \quad \forall u, v$$

$$\Rightarrow \langle u, (S_1 - S_2)(v) \rangle \quad \forall u, v$$

$$\Rightarrow \langle (S_1 - S_2)(u), (S_1 - S_2)(v) \rangle = 0 \quad \forall v$$

$$\Rightarrow S_1(v) = S_2(v) \quad \forall v \Rightarrow S_1 = S_2.$$

Adjoint of T is denoted by T^* .

Existence: Let V be a f.d.i.p.s. $T \in L(V)$ then $\exists T^*$ s.t.

$$\langle T(u), v \rangle = \langle u, T^*(v) \rangle. \quad \forall u, v \in V.$$

Let $T \in L(V)$.

define $f: V \rightarrow \mathbb{F}$ s.t.

$$f(v) = \langle T(v), u \rangle \quad \forall v \in V \quad \text{--- } (*)$$

$$\Rightarrow f \in V'$$

$$\exists u_f \in V \quad \text{s.t.} \quad f(v) = \langle v, u_f \rangle \quad \forall v \in V \quad \text{--- } (**)$$

$$\text{from } (*) \text{ and } (**) \Rightarrow \langle T(v), u \rangle = \langle v, u_f \rangle \quad \forall v \in V \quad \text{--- } (***)$$

$$\Rightarrow T^*(u) = u_f.$$

define $T^*: V \rightarrow V$ s.t. $T^*(u) = u_f. \forall u \in V$

from ~~(*)~~ $\langle T(u), u \rangle = \langle u, T^*(u) \rangle$

Exercise: show $T^* \in L(V)$.

V f.d.i.p.s.

$T: V \rightarrow V$, L.T.

$\beta = \{e_1, \dots, e_n\}$ orthonormal basis.

$$[T]_{\beta} = [a_{ij}]. \text{ then } a_{ij} = \langle T(e_j), e_i \rangle \forall i, j$$

$$\text{Pf: } T(e_j) = \sum_{k=1}^n a_{kj} e_k$$

$$\langle T(e_j), e_i \rangle = a_{ij}$$

Ex.: V f.d.i.p.s.

$T \in L(V)$

$B = \{e_1, \dots, e_n\}$ is orthonormal basis

$$[T^*]_B = [T]_B^*$$

$A_{n \times n}$ over $\mathbb{C}$

unitary $A^* A = I$

normal $A^* A = A A^*$

self adjoint $A^* = A$.

$T \in L(V)$ $\left\{ \begin{array}{l} T \text{ is normal if } T^* T = T T^* \\ T \text{ is unitary if } T^* T = I \\ T \text{ is self adj if } T^* = T \end{array} \right.$

$\Leftrightarrow$ Matrix rep. follow same conditions.

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$V \leftarrow \text{f.l.v.s}$

$T: V \rightarrow V$

β, β' are basis of V

$\begin{matrix} \parallel & \searrow \\ \{v_1, \dots, v_n\} & \{u_1, \dots, u_n\} \end{matrix}$

$$\exists P \text{ s.t. } P^{-1} [T]_{\beta} P = [T]_{\beta'}$$

$$v_i = \sum_{j=1}^n p_{ij} u_j$$

$$P = [p_{ij}]$$

A and B are similar

if $\exists P$ s.t. $P^{-1} A P = B$.

$(M_{n \times n}(\mathbb{C}), \sim)$

$\uparrow$ similarity relation
(Equivalence relation)

$$x \in M_{n \times n}(\mathbb{C}) \quad , \quad E_x = \{y \in M_{n \times n}(\mathbb{C}) : x \sim y\}$$

$$M_{n \times n} = \bigsqcup_x E_x.$$

$$A \sim B$$

Then trace, det, rank, eig. values, Min. poly., char all same.

So for each eq. class find easiest to work w/ simplest matrix.

Invariant subspace: $V \leftarrow \text{V.S.}$

$T \in L(V)$, $U \subseteq V$ subspace

Then U is called T invariant / inv. under T
if $T(U) \subseteq U$.

obs. : $V \leftarrow \text{f.d.v.s.}$

U subspace of V inv. under $T \in L(V)$
 $\{v_1, \dots, v_k\}$ basis of U .

$\beta = \{v_1, \dots, v_k, u_1, \dots, u_{n-k}\}$ basis of V .

$$[T]_{\beta} = \left[\begin{array}{c|c} A_{1 \times k} & A_2 \\ \hline 0 & A_3 \end{array} \right]$$

$U_1, U_2 \subseteq V$ T inv. s.t. $V = U_1 \oplus U_2$.

$\beta = \{ \underbrace{v_1, \dots, v_n}_{\text{Basis of } U_1}, \underbrace{u_1, \dots, u_{n-n}}_{\text{Basis of } U_2} \}$

$$[T]_{\beta} = \left[\begin{array}{c|c} A_1 & 0 \\ \hline 0 & A_2 \end{array} \right]$$

(iii) $V = U_1 \oplus U_2 \oplus U_3$

$$\left[\begin{array}{c|c|c} A_1 & 0 & 0 \\ \hline 0 & A_2 & 0 \\ \hline 0 & 0 & A_3 \end{array} \right]$$

Suppose U_i $i=1, 2, \dots, n$ are T -inv. subspaces
s.t. $V = \bigoplus_{i=1}^n U_i$, $\dim(U_i) = 1$.

$$U_i = \text{span}(u_i)$$

consider $\beta = \{u_1, \dots, u_n\}$ Basis

$$[T]_{\beta} = \left[\begin{array}{c} \diagup 0 \\ 0 \end{array} \right] \text{ diag. matrix.}$$

Result: Let $V \leftarrow \text{f.d.v.s}$ $T \in L(V)$.

$U \subseteq V$ and U is T inv.

then $U = \text{span}(u), u \in U \Leftrightarrow T(u) = \alpha u$ for some $\alpha \in F$

Defⁿ: Let $T \in L(V)$ and let $\exists u \in U, u \neq 0$

$$T(u) = \lambda u, \lambda \in F.$$

then λ is called eigenvalue of T
 u is called ^{corresponding} eigenvector

$\text{eig}(T) \leftarrow$ set of all eigenvalues of T .

$$\begin{aligned} \lambda \in \text{eig}(T) &\Rightarrow \exists u \neq 0 \text{ s.t. } T(u) = \lambda u \\ &\Rightarrow (T - \lambda I)u = 0 \\ &\Rightarrow u \in \text{Null}(T - \lambda I) \end{aligned}$$

$\text{Null}(T - \lambda I)$ is called the eigenspace of T w.r.t. λ eigenvalue.

$$A \in M_{m \times m}(\mathbb{C}).$$

$\lambda \in \mathbb{C}$ is called an eigenvalue of A if $\exists x \in \mathbb{C}^n$
s.t. $Ax = \lambda x$.

$\text{eig}(A) \leftarrow$ set of eigenvalues of A .

$$\lambda \in \text{eig}(A) \Leftrightarrow \exists x \neq 0 (A - \lambda I)x = 0 \Leftrightarrow x \in \underbrace{\text{Null}(A - \lambda I)}_{\text{eigenspace of } A \text{ for } \lambda}$$

$$\lambda \in \text{eig}(A) \Leftrightarrow \exists x \neq 0, (A - \lambda I)x = 0 \Leftrightarrow \det(A - \lambda I) = 0.$$

$$p_A(\lambda) = \det(A - \lambda I) \rightsquigarrow \text{characteristic polynomial of } A$$

$$p_A(\lambda) = 0 \text{ is characteristic equation.}$$

roots = eigenvalue.

Result: $V \leftarrow \text{f.d.v.s}$, $T \in L(V)$, β be any basis of V

then $\text{eig}(T) = \text{eig}([T]_\beta)$.

Exercise 1, p. 11 of
(idea of proof)

$$\lambda \in \text{eig}(T) \Rightarrow T(v) = \lambda v \Rightarrow [T(v)]_\beta = [\lambda v]_\beta \Rightarrow [T]_\beta [v]_\beta = \lambda [v]_\beta$$

def: $T \in L(V)$, $\det(T) = \det([T]_B)$. B is any basis

· Characteristic polynomial $p_T(\lambda) = \det(T - \lambda I) = \det([T - \lambda I]_{\mathcal{B}})$

Does every A has an eigen value?

No $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ over $\mathbb{R}$, $p_A(\lambda) = \lambda^2 + 1$
no root over $\mathbb{R}$

Yes $\begin{bmatrix} 0 & -1 \\ 1 & 1 \end{bmatrix}$ over $\mathbb{C}$ has eigen values.